Statistical Analysis and Hypothesis Testing Report

Project Overview

This project performs statistical analysis on sales data to understand business performance, identify trends, evaluate relationships between variables, and generate actionable recommendations.

Objectives

• Calculate descriptive statistics
• Analyze data distributions
• Measure correlations
• Perform hypothesis testing
• Compute confidence intervals
• Develop regression models
• Generate business insights

Dataset Description

The dataset contains Date, Product, Quantity, Price, Customer_ID, Region, and Total_Sales fields.

Descriptive Statistics

Mean, Median, Mode, Standard Deviation, and Variance were calculated to summarize sales performance.

Distribution Analysis

A histogram and Shapiro-Wilk normality test were used to analyze data distribution.

Correlation Analysis

Pearson correlation coefficients and a heatmap were generated to study relationships among variables.

Hypothesis Testing

Independent T-Test and One-Way ANOVA were performed to compare regional sales performance.

Confidence Intervals

A 95% confidence interval and margin of error were calculated for Total Sales.

Regression Analysis

A linear regression model was built using Quantity and Price to predict Total Sales. R-squared was used for evaluation.

Business Recommendations

Increase marketing for top-performing products, expand in strong regions, improve customer retention, monitor low-performing products, and use predictive analytics for planning.

Conclusion

The project demonstrates a complete statistical analysis workflow and supports data-driven business decisions.